



Juergen Gall
Introduction
Computer Vision
MA-INF 2201 / MA-MOROB-M04
WS25/26

Organization

- Course if offered for
 - Computer Science (M.Sc.)
 - Mobile Robotics (M.Sc.)
 - Erasmus students

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Organization

- Lecturer:
 - Prof. Dr. Juergen Gall
 - https://pages.iai.uni-bonn.de/gall_juergen
- Teaching:

Prof. Dr. Juergen Gall
Research Interests
Job Offers/Theses
Publications
Software
Data
Projects
Conferences
Teaching

Teaching

PhD Seminar - Graphics, Vision, Audio for Intelligent Systems, SS23, WS23/24, SS24, WS24/25, SS25, WS25/26, SS26, WS26/27, SS27, WS27/28, SS28, WS28/29, SS29, WS29/30
 BA-INF 062 - Begleitseminar zur Bachelorarbeit: Computer Vision, SS24, WS24/25, SS25, WS25/26, SS26, WS26/27, SS27, WS27/28, SS28, WS28/29, SS29, WS29/30, SS30, WS30/31, SS31, WS31/32, SS32, WS32/23, SS23, WS23/24, SS24, WS24/25, SS25, WS25/26
 MA-INF 2201/MA-MOROB-M04 - Computer Vision, WS23/24, WS24/25, WS25/26, WS26/27, WS27/28, WS28/29, WS29/30, WS30/31, WS31/32, WS32/23, WS23/24, WS24/25, WS25/26
 MA-INF 2203 - Advanced Computer Vision (Computer Vision II), SS24, SS25, SS26, SS27, SS28, SS29, SS30, SS31, SS32, SS23, SS24, SS25

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Organization

- Slides:
 - https://pages.iai.uni-bonn.de/gall_juergen/teaching/Lectures/cv25.html

MA-INF 2201/MA-MOROB-M04 - Computer Vision

2L + 2E, WS25/26

Lecturer

Juergen Gall

Degree programs

The lecture is offered for the Master of Science degrees: Computer Science (MA-INF 2201) and Mobile Robotics (MA-MOROB-M04).

Content

Tentative: Linear filters, Edges, Derivatives, Hough Transform, Segmentation, Graph Cuts, Mean Shift, Active Contours, Level Sets, MRFs, Expectation Maximization, Temporal Filtering, Active Appearance Models, Shapes, Optical Flow, 2D Tracking, Camera, 2D/3D Features, Stereo, 3D Reconstruction, 3D Pose Estimation, A Deformable Meshes, RGBD Vision

Material

The slides and recordings are available at Slides/Recordings.

Prerequisites

Basic knowledge of linear algebra, analysis, probability theory, Python programming

Schedule

Tuesday, 10:15-11:45, HS XVI, Nußallee 17

Friday, 10:15-11:45, HS XVI, Nußallee 17

Start: Tuesday, 14.10.

Organization

- Slides via sciebo:
 - <https://hochschulcloud.nrw/en/index.html>



- Password: MA-INF2201

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Organization

- Structure: 4+2 SWS
- Lecture:
 - HS XVI, Nußallee 17
 - Tuesday, 10:15-11:45
 - Friday, 10:15-11:45
 - Lectures will be recorded (zoom link will be sent)

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Organization



- Structure: 4+2 SWS
- Exercise:
 - Room 2.025
Institute of Computer Science
Friedrich-Hirzebruch-Allee 8
 - Start: 22.10.
 - Wednesday, 12:00-13:00
 - Wednesday, 13:00-14:00
 - Wednesday, 14:00-15:00
 - Wednesday, 15:00-16:00

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Organization



- Feedback welcome!
- There is time to repeat parts of previous lecture (only on request)
- Two recap lectures
- No explicit exam preparation (attend lecture and solve exercise sheets)
- Written exam
- Contact per email:
 - gall@iai.uni-bonn.de
 - Email with subject: [MA-INF 2201]

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Exercise



- Small implementation exercises
- Exercises are not exam questions, but understanding → Able to implement it
- No programming skills → No job as computer scientist
- OpenCV and Python (<http://opencv.org/>)

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Exercise



- You can use Claude, Github Copilot, Gemini Code Assist...
...but you need to understand your code.
- Remember: In a job interview you need to program without these tools
- If the tutors have some doubts they might ask you to explain your solution.
- Quality of code is relevant for grading

At least 50% of the points are required to qualify for exam

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Organization



- Registration for exercise today!

Exercises

Theory and programming. At least 50% of the exercise points are required to qualify for exam. The programming exercises are implemented in PyTl implementing algorithms presented in the lecture. If you are not comfortable with OpenCV, we recommend to have a look at the OpenCV Tutorials. OpenCV will be released on the date of the first lecture.

Schedule

Wednesday, 12:00-13:00, 2.025, Institute of Computer Science, Friedrich-Hirzebruch-Allee 8
 Wednesday, 13:00-14:00, 2.025, Institute of Computer Science, Friedrich-Hirzebruch-Allee 8
 Wednesday, 14:00-15:00, 2.025, Institute of Computer Science, Friedrich-Hirzebruch-Allee 8
 Wednesday, 15:00-16:00, 2.025, Institute of Computer Science, Friedrich-Hirzebruch-Allee 8
 Start: Wednesday, 22.10.

Register until 14.10. for the exercise [registration form](#). For the registration, you have to use your email address of the University of Bonn. If you did not manage to register until 14.10. please contact us until 15.10. as soon as possible.

- Use email address of sciebo account (university email)

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Organization

- Use email address of sciebo account (university email)
- There will be four time slots for the first exercise

First Name *

Meine Antwort

Family Name *

Meine Antwort

Sciebo Username / Uni Email *

Meine Antwort

Degree / Major *

☐ Computer Science

☐ Mobile Robotics

☐ Sonstiges:

Which time slot of the exercise is possible for you to attend? *

☐ 12:00-14:00

☐ 14:00-16:00

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Sciebo



- <https://hochschulcloud.nrw/en/index.html>
- Registration / Select University

To register for sciebo or to change your settings, select your institution. You will then be taken to your organisation's login page.

Universität Bonn

Auswählen / Select

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Sciebo



- <https://hochschulcloud.nrw/en/index.html>
- Registration / Select University
- Login with your university email



Login to sciebo - the Campuscloud

Forgot your password?

Uni-ID

Password

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Organization



- Registration for exercises today
- Submission in groups of up to 2 persons allowed
- Submission deadline Sunday midnight
- First exercise sheet is in the slides folder
- Submission for first sheet is Sunday, 19.10., midnight
- How to submit: You will get details per email after registration

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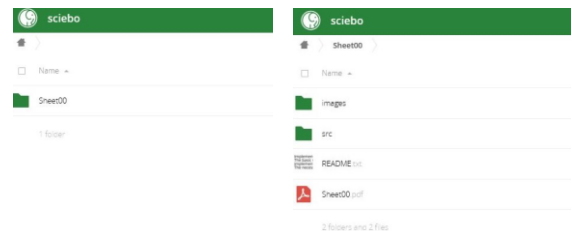
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Organization



- Exercise sheets via sciebo (<https://hochschulcloud.nrw/en/index.html>):



- Create sciebo account (university email)

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Organization



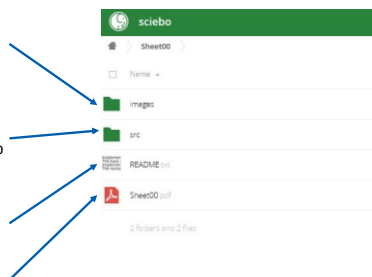
- Exercise sheets via sciebo:

Data
(do not upload for submission)

Template
(modify and submit only template and readme; do not change paths)

Readme
(add group members)

Exercise Sheet



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Exercises



- Use Python and OpenCV (versions are specified in first exercise sheet)
- Submit README.txt with your group members for all exercise sheets (full names)
- Only one submission per group
- Submit modified template

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Exercises



- **Never submit code that does not run** (zero points)
- Add comments to the code
- Use readable variable and function names (e.g. `opticalflow_x` instead of `a`)
- Points can be removed per sheet for unreadable programming style

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Exercises



Paper exercise:

- Submit as single PDF
- Hand writing → Photo → Image → Save as PDF
- You can also use Latex or other tools to create PDF

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Exercises



- Current exercise sheet 0 is a bonus sheet
- We start with weekly exercise sheets (tentative first three exercise sheet, 20 points each)
- We continue later with biweekly sheets (tentative 6 exercise sheets, 40 points each)

At least 50% of the points are required to qualify for exam

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Registration for Exercise groups



If you have troubles to register or to submit, please contact
 Jinhui Yi
yij@iai.uni-bonn.de

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Mailinglist



- After registration, you will be added to a mailinglist
- You can unsubscribe at any time

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Exam



- Written exam
- Tentative:
 - 1st Exam: Thursday, 24.2. (tbc)
 - 2nd Exam: Monday, 20.3. (tbc)
- **Important: There will be no exams in summer semester**
- If you pass the exercises, you do not need to take the exercises again.

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Registration



- Today via https://pages.iai.uni-bonn.de/gall_juergen/teaching/Lectures/cv25.html for setting up sciebo and assigning exercise groups (**This is not an official registration**)
- Registration for exercise and lecture: **13.10.-31.10.** <https://basis.uni-bonn.de>
- Registration for exam: 1.12.-21.12. <https://basis.uni-bonn.de>
- **There might be other dates for Mobile Robotics**

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How to attend a lecture



- Attend lecture
- After lecture go through the slides on the same or next day (2-3 hours)
- If something is unclear, check recording / additional reading material
- If something is still unclear write me an email and ask to repeat part of previous lecture
- **Try to solve all exercise sheets**

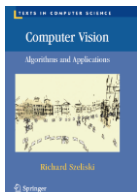
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Literature



Basic material:

- <http://szeliski.org/Book/>
- <http://www.robots.ox.ac.uk/~vgg/hzbook/>



Computer Vision - Algorithms and Applications, Szeliski, Richard, Springer, 2011



Multiple View Geometry in Computer Vision, Second Edition, Richard Hartley and Andrew Zisserman, Cambridge University Press, 2004.

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Literature



Basic material:

- <http://web4.cs.ucl.ac.uk/staff/s.prince/book/book.pdf>



S. Prince. Computer Vision: Models, Learning, and Inference. Cambridge University Press 2012

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Literature



Additional material as references on slides.

Main conferences:

- IEEE Conference on Computer Vision and Pattern Recognition (CVPR)
- IEEE International Conference on Computer Vision (ICCV)
- European Conference on Computer Vision (ECCV)

Main journals:

- IEEE Transactions on Pattern Analysis and Machine Intelligence
- International Journal of Computer Vision

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Impact of Computer Vision



- Impact of CVPR is comparable to Nature or Science

Publication	h5-Index	h5-Medianwert
1. Nature	488	745
→ IEEE/CVF Conference on Computer Vision and Pattern Recognition	440	689
3. The New England Journal of Medicine	434	897
4. Science	409	633
5. Nature Communications	375	492
6. The Lancet	368	678
→ Neural Information Processing Systems	337	614
8. Advanced Materials	327	420
9. Cell	320	482
→ International Conference on Learning Representations	304	584

[https://scholar.google.com/citations?view_op=top_venues]

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Impact of Computer Vision

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- CVPR is in the top 5 of scientific journals or proceedings with highest impact
- Most important field in computer science: Computer Vision
- Second most important field in computer science: Machine Learning

[https://scholar.google.com/citations?view_op=top_venues]

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What is Computer Vision?

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Convert visual data into “meaningful” numbers



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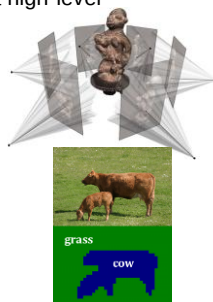
22

What are “meaningful” numbers?

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From low-level (pixel values) to a high-level representation:

- Metric measurements
- Models
- Labels



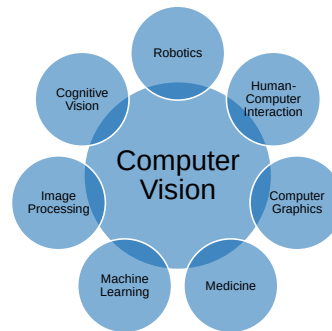
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Connections to other disciplines

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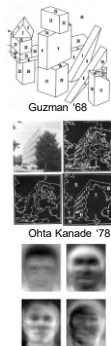
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Brief history of computer vision

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1960's: interpretation of synthetic worlds
 1970's: some progress on interpreting selected images
 1980's: shift toward geometry and increased mathematical rigor
 1990's: face recognition; statistical analysis in vogue
 2000's: broader recognition; large annotated datasets available; video processing starts
 2010's: deep learning; many commercial applications and successful spin-offs
 2020's: vision language models



Source: J. Hays

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Inherent ambiguity of the problem

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Many different 3D scenes could have given rise to a particular 2D picture

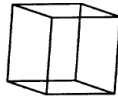


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Source: S. Lazebnik

A simple idea to recover 3D shapes from line drawings



Task: Given the set of image coordinates of the vertices (x_i, y_i) , recover the world coordinates (X_i, Y_i, Z_i) .

We will assume orthographic projection so that $X_i = x_i$ and $Y_i = y_i$

Then, the problem is: recover the missing depth Z_i

Marill, AI-Memo-1136, 1989

Source: A. Torralba

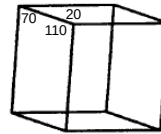
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A simple idea to recover 3D shapes from line drawings



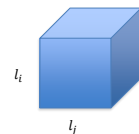
Heuristic: find Z_k that minimize the standard deviation of angles in the 3D object

Possible solution 1



$$Z_k = 0 \text{ for all } k$$

Possible solution 2



All 90 degree angles

Minimize standard deviations of angles: $P_k = (X_k, Y_k, Z_k)$

$$\argmin_k \frac{1}{N-1} \sum_{i < j} \left(\alpha_{ij} - \frac{1}{N} \sum_{i < j} \alpha_{ij} \right)^2$$

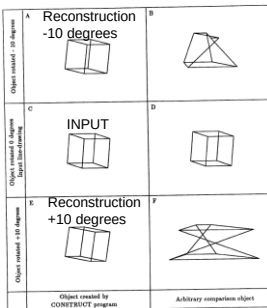
$$\alpha_{ij} = \cos^{-1}(\mathbf{u}_i \cdot \mathbf{u}_j)$$

$$\text{unit length vectors: } \mathbf{u}_i = \frac{l_i}{\|l_i\|}$$

Source: A. Torralba

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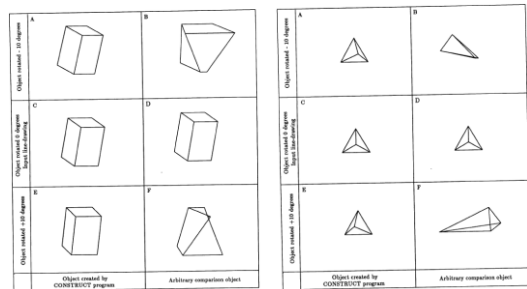
A simple idea to recover 3D shapes from line drawings



Source: A. Torralba

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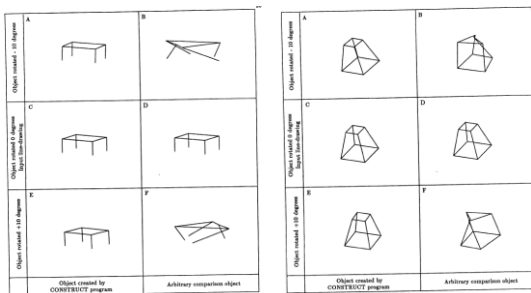
A simple idea to recover 3D shapes from line drawings



Source: A. Torralba

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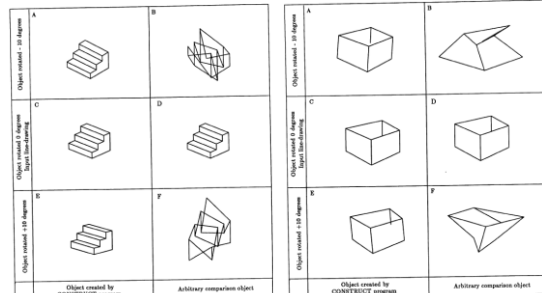
A simple idea to recover 3D shapes from line drawings



Source: A. Torralba

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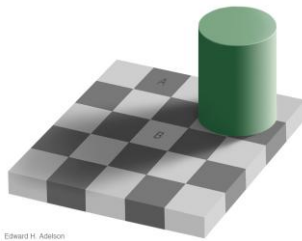
A simple idea to recover 3D shapes from line drawings



Source: A. Torralba

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Measuring light vs. measuring scene properties



Edward H. Adelson

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Source: A. Torralba

Measuring light vs. measuring scene properties



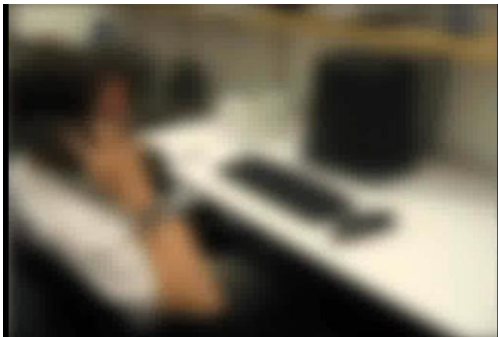
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Source: A. Torralba

Context



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Source: A. Torralba

Humans work with strong priors



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Source: A. Torralba

Conclusion



- Human vision relies not **only** on data term (pixel information)
- Priors are important
- They explain the **most likely** scene if data is ambiguous:

$$\operatorname{argmax}_y P(y|I) = \operatorname{argmax}_y P(I|y)P(y)$$

- Bayes formula:

$$P(y|I) = \frac{P(I|y)P(y)}{P(I)}$$

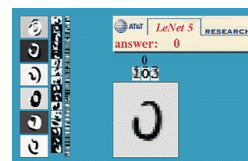
Posterior = Likelihood x Prior

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Optical character recognition (OCR)



Digit recognition



License plate readers



Automatic check processing

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Source: S. Seitz

Face detection

Digital cameras detect faces



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Smile detection

The Smile Shutter flow

Imagine a camera smart enough to catch every smile! In Smile Shutter Mode, your Cyber-shot® camera can automatically trip the shutter at just the right instant to catch the perfect expression.



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Image classification



<http://www.clarifai.com/>

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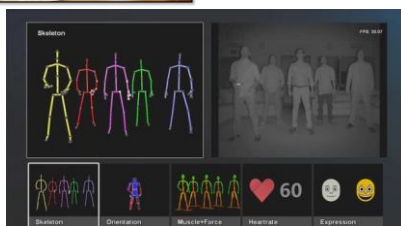
Automotive safety

Pedestrian collision warning, forward collision warning, lane departure warning...



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Vision-based interaction: Xbox Kinect



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Movie production



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Generative AI

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Vision for robotics, space exploration

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NASA's Mars Exploration Rover Spirit captured this westward view from atop a low plateau where Spirit spent the closing months of 2007.

Vision systems (JPL) used for several tasks

- Panorama stitching
- 3D terrain modeling
- Obstacle detection, position tracking
- Matthies et al. Computer Vision on Mars IJCV'07

Source: S. Seitz

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Jobs, jobs, ...

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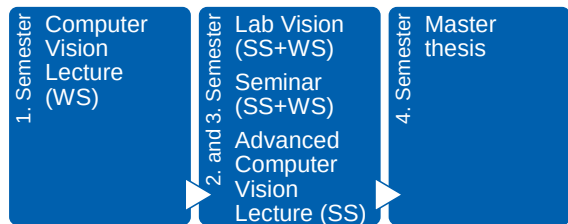
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Curriculum

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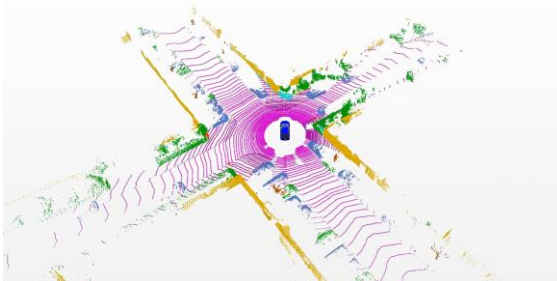
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Semantic Segmentation

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[J. Behley et al. SemanticKITTI: A Dataset for Semantic Scene Understanding of LIDAR Sequences, ICCV 2019]

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Human Pose Estimation

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[A. Doering et al. Joint Flow: Temporal Flow Fields for Multi Person Tracking, BMVC 2018]

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Video Understanding



Video Captioning Task Results



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Video Question Answering



Video-Panda

Demos from TGIF-QA

[J. Yi et al. Video-Panda:
Parameter-efficient
Alignment for Encoder-free
Video-Language Models.
CVPR 2025]

20.04.2024

Research Unit 2335 - Anticipating Human Behavior

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Action Segmentation



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Anticipating Actions



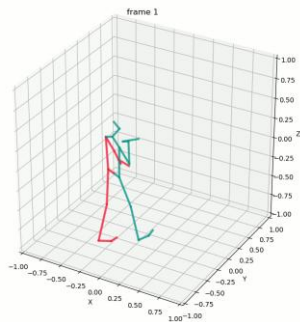
[Y. Abu Farha et al. When will you do what?
Anticipating Temporal Occurrences of Activities. CVPR 2018]

20.02.2019

Research Unit 2335 - Anticipating Human Behavior

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Forecast Human Pose



[Ruiz et al. Human Motion Prediction via Spatio-Temporal Inpainting. ICCV 2019]

23.09.2019

Research Unit 2335 - Anticipating Human Behavior

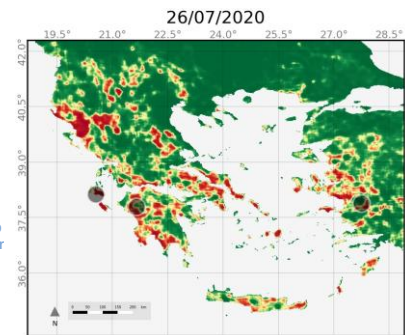
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Wild Fire Forecasting



[M.H.S. Eddin et al.
Location-Aware
Adaptive
Normalization: A Deep
Learning Approach for
Wildfire Danger
Forecasting. IEEE
Transactions on
Geoscience and
Remote Sensing 2023]

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Cluster of Excellence - PhenoRob

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[<http://www.phenorob.de>]

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Organization

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- Slides: Password: MA-INF2201
- Registration for exercises today
- Registration of sciebo account
- Submission for first sheet (5 bonus points) is Sunday, 19.10.

13.10.2025

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Thank you for your attention.



11.10.2025

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